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QUANTITATIVE APTITUDE

Quantitative Aptitude & Numerical Ability Master E-Book

Complete Mathematical Concepts, Formulas, Shortcuts & Solved Examples

Target Role: Infosys SE (2026 Batch)

Total Questions: 90 Solved MCQs

Format: CampusMonk Master E-Book Standard

Status: 100% Verified & Answer-Keyed

Q001 APTITUDE (TECHNICAL ABILITY)

If a:b = 3:4 and b:c = 5:6, find a:b:c.

A 3:4:6

B 15:20:24

C 15:20:30

D 3:5:6

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

Make b the same in both ratios. LCM of 4 and 5 is 20. Scale a:b = 3:4 by 5 → 15:20. Scale b:c = 5:6 by 4 → 20:24. Combined ratio a:b:c = 15:20:24.

Q002 APTITUDE (TECHNICAL ABILITY)

A train 180 m long is running at 54 km/h. How long will it take to pass a pole?

A 10 seconds

B 12 seconds

C 15 seconds

D 18 seconds

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

Speed = 54 × 5/18 = 15 m/s. Time to pass a pole = length / speed = 180 / 15 = 12 seconds.

Q003 APTITUDE (TECHNICAL ABILITY)

Find the remainder when 3^25 is divided by 5.

A 1

B 2

C 3

D 4

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Powers of 3 modulo 5 cycle every 4: 3, 4, 2, 1. Since 25 = 4×6 + 1, the remainder is the 1st residue in the cycle, which is 3.

Q004 APTITUDE (TECHNICAL ABILITY)

Find the compound interest on ₹5000 for 2 years at 10% per annum, compounded annually.

A ₹1000
B ₹ 1050
C ₹1100
D ₹1200

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

CI = P[(1 + R/100)^n − 1] = 5000[(1.10)^2 − 1] = 5000 × 0.21 = ₹1050.

Q005 APTITUDE (TECHNICAL ABILITY)

A can finish a work in 15 days and B can finish the same work in 10 days. In how many days will they finish the work if they work together?

A 5 days
B 6 days
C 7.5 days
D 8 days

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

A's 1-day work = 1/15, B's 1-day work = 1/10. Together = 1/15 + 1/10 = 1/6. So they finish the work in 6 days.

Q006 APTITUDE (TECHNICAL ABILITY)

Two fair dice are thrown together. What is the probability of getting a sum of 7?

A 1/12
B 1/8
C 1/6
D 5/36

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) — 6 outcomes. Total outcomes = 36. Probability = 6/36 = 1/6.

Q007 APTITUDE (TECHNICAL ABILITY)

A number is first increased by 20% and then decreased by 20%. What is the net percentage change?

(A) No change
(B) 4% increase
(C) 4% decrease
(D) 2% decrease

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Net multiplying factor = 1.20 × 0.80 = 0.96. This is a 4% decrease from the original value.

Q008 APTITUDE (TECHNICAL ABILITY)

A shopkeeper marks an article 25% above the cost price and then gives a discount of 10%. Find his profit percentage.

(A) 10%
(B) 12%
(C) 12.5%
(D) 15%

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Take CP = 100. Marked price = 125. After 10% discount, SP = 125 × 0.90 = 112.5. Profit = 12.5%.

Q009 APTITUDE (TECHNICAL ABILITY)

The length of a rectangle is increased by 20% and the breadth is decreased by 10%. What is the percentage change in its area?

(A) 8% increase
(B) 8% decrease
(C) 10% increase
(D) No change

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
New area factor = 1.20 × 0.90 = 1.08.
The area increases by 8%.

Q010 APTITUDE (TECHNICAL ABILITY)

The HCF and LCM of two numbers are 18 and 252 respectively. If one number is 36, find the other number.

A 108
B 126
C 144
D 162

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

Product of the two numbers = HCF × LCM = 18 × 252 = 4536. Other number = 4536 / 36 = 126.

Q011 APTITUDE (TECHNICAL ABILITY)

A can finish a job in 12 days and B can finish the same job in 18 days. How many days will they take working together?

A 6 days
B 7.2 days
C 8 days
D 9 days

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

A's 1-day work = 1/12. B's 1-day work = 1/18. Together = 1/12 + 1/18 = 3/36 + 2/36 = 5/36. Time = 36/5 = 7.2 days.

Q012 APTITUDE (TECHNICAL ABILITY)

A trader sells an item at 8% loss. If he had sold it for ₹96 more, he would have gained 4%. What is the cost price of the item?

A ₹700
B ₹750
C ₹800
D ₹850

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Let CP = x. First SP = 0.92x. Second SP = 1.04x. Difference = 0.12x = 96, so x = 96 / 0.12 = 800. Cost price is ₹800.

Q013 APTITUDE (TECHNICAL ABILITY)

A team of 3 is to be chosen from 5 developers and 4 testers. In how many ways can the team be formed if it must include at least 1 tester?

A 64
B 70
C 74
D 84

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Total ways to choose 3 from 9 = C(9,3) = 84.
Ways with no tester = C(5,3) = 10. So at least one tester = 84 − 10 = 74.

Q014 APTITUDE (TECHNICAL ABILITY)

In how many distinct ways can the letters of the word SERVER be arranged?

A 120
B 180
C 240
D 720

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
SERVER has 6 letters with R and E repeating twice each. Number of distinct arrangements = 6! / (2! × 2!) = 720 / 4 = 180.

Q015 APTITUDE (TECHNICAL ABILITY)

An article is bought for ₹800 and sold for ₹920. What is the profit percentage?

A 12%
B 15%
C 18%
D 20%

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Profit = 920 − 800 = 120. Profit % = (120 / 800) × 100 = 15%.

Q016 APTITUDE (TECHNICAL ABILITY)

A and B together can do a piece of work in 10 days. A alone can do it in 15 days. In how many days can B alone do the same work?

A 20 days
B 25 days
C 30 days
D 45 days

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

A + B = 1/10 and A = 1/15. B = 1/10 − 1/15 = 3/30 − 2/30 = 1/30. So B alone takes 30 days.

Q017 APTITUDE (TECHNICAL ABILITY)

The marked price of a laptop is ₹40,000. After a discount of 10%, it is sold. If the cost price is ₹32,000, what is the profit percentage?

A 10%
B 12.5%
C 15%
D 20%

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

SP after 10% discount = 90% of 40,000 = ₹36,000. Profit = 36,000 − 32,000 = ₹4,000. Profit % = (4000 / 32000) × 100 = 12.5%.

Q018 APTITUDE (TECHNICAL ABILITY)

How many 4-letter codes can be formed from the letters of the word ORANGE if no letter is repeated?

A 120
B 240
C 360
D 720

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

ORANGE has 6 distinct letters. Number of 4-letter codes with no repetition is P(6,4) = 6×5×4×3 = 360.

Q019 APTITUDE (TECHNICAL ABILITY)

A bag contains 4 red and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both are red?

A 2/15
B 1/5
C 4/15
D 2/9

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
P = C(4,2) / C(10,2) = 6 / 45 = 2/15.
Sequentially: (4/10)×(3/9) = 12/90 = 2/15.

Q020 APTITUDE (TECHNICAL ABILITY)

Two fair dice are thrown. What is the probability that the sum is 9?

A 1/12
B 1/9
C 1/8
D 1/6

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Favourable pairs: (3,6), (4,5), (5,4), (6,3)
— 4 outcomes. Total outcomes = 36. Probability = 4/36 = 1/9.

Q021 APTITUDE (TECHNICAL ABILITY)

The value of 40% of 250 + 15% of 400 is

A 140
B 150
C 160
D 170

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
40% of 250 = 100. 15% of 400 = 60. Sum = 160.

Q022 APTITUDE (TECHNICAL ABILITY)

Pipe A fills a tank in 10 hours and pipe B empties it in 15 hours. If both are opened together, the tank fills in

A 20 hours
B 25 hours
C 30 hours
D 6 hours

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Net work per hour = 1/10 − 1/15 = 1/30. Time = 30 hours.

Q023 APTITUDE (TECHNICAL ABILITY)

The average of five numbers is 18. If one number 22 is replaced by 12, the new average is

A 16
B 17
C 15
D 14

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
Total was 90. New total = 90 − 22 + 12 = 80. New average = 80/5 = 16.

Q024 APTITUDE (TECHNICAL ABILITY)

The present ages of a father and son are in the ratio 7 : 2. After 6 years the ratio will be 8 : 3. The son's present age is

A 10 years
B 12 years
C 14 years
D 16 years

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Let ages be 7x and 2x. (7x+6)/(2x+6) = 8/3. 3(7x+6)=8(2x+6) → 21x+18=16x+48 → 5x=30 → x=6. Son = 12.

Q025 APTITUDE (TECHNICAL ABILITY)

A invests ₹30,000 for 8 months and B invests ₹40,000 for 6 months. If the profit is ₹5,500, A's share is

(A) ₹2,750
(B) ₹2,200
(C) ₹3,000
(D) ₹3,300

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
Ratio of capitals × time = 30,000×8 : 40,000×6 = 240 : 240 = 1 : 1. A gets half of 5,500 = ₹2,750.

Q026 APTITUDE (TECHNICAL ABILITY)

₹5,000 is lent at 8% simple interest per year. What is the interest after 3 years?

(A) ₹800
(B) ₹1,000
(C) ₹1,200
(D) ₹1,500

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
SI = PRT/100 = 5000×8×3 / 100 = ₹1,200.

Q027 APTITUDE (TECHNICAL ABILITY)

If A : B = 3 : 4 and B : C = 6 : 7, then A : B : C is

(A) 3 : 4 : 7
(B) 9 : 12 : 14
(C) 18 : 24 : 14
(D) 3 : 6 : 7

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
B must match: A:B = 3:4 = 9:12 and B:C = 6:7 = 12:14. So A:B:C = 9:12:14.

Q028 APTITUDE (TECHNICAL ABILITY)

A number is increased by 25% and then decreased by 20%. The net change is

(A) 5% increase
(B) 5% decrease
(C) no change
(D) 2.5% decrease

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Let the number be 100. After +25% it is 125. After −20% it is 100. Net change is zero.

Q029 APTITUDE (TECHNICAL ABILITY)

What is the remainder when 2^10 is divided by 7?

(A) 1
(B) 2
(C) 3
(D) 4

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
2^1=2, 2^2=4, 2^3=1 (mod 7), then cycle length 3. 10 = 3×3 + 1, so remainder equals 2^1 = 2.

Q030 APTITUDE (TECHNICAL ABILITY)

A train 180 m long runs at 54 km/h. How many seconds does it take to pass a pole?

(A) 10 s
(B) 12 s
(C) 15 s
(D) 18 s

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
54 km/h = 54×5/18 = 15 m/s. Time = 180/15 = 12 s.

Q031 APTITUDE (TECHNICAL ABILITY)

The sum of ages of A and B is 40. A is 4 years older than B. B's age is

A 16
B 18
C 20
D 22

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

A=B+4
and A+B=40 → 2B+4=40 → B=18.

Q032 APTITUDE (TECHNICAL ABILITY)

An article bought for ₹450 is sold at 12% loss. The selling price is

A ₹ 396
B ₹ 400
C ₹ 405
D ₹ 414

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

SP
= 88% of 450 = ₹396.

Q033 APTITUDE (TECHNICAL ABILITY)

Divide ₹1,260 in the ratio 2 : 3 : 4. The middle share is

A ₹ 280
B ₹ 360
C ₹ 420
D ₹ 560

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Parts
= 9. One part = 140. Middle = 3×140 = ₹420.

Q034 APTITUDE (TECHNICAL ABILITY)

A
shopkeeper marks goods 40% above cost and gives a 10% discount.
His profit percent is

(A) 24%
(B) 26%
(C) 30%
(D) 32%

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Let
CP=100. MP=140. SP=126. Profit=26%.

Q035 APTITUDE (TECHNICAL ABILITY)

The
HCF of 48 and 180 is

(A) 6
(B) 8
(C) 12
(D) 24

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
48=2^4×3,
180=2^2×3^2×5. HCF = 2^2×3 = 12.

Q036 APTITUDE (TECHNICAL ABILITY)

A
car covers 240 km in 4 hours. Its speed in m/s is

(A) 15 m/s
(B) 16.67 m/s
(C) 20 m/s
(D) 60 m/s

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Speed
= 60 km/h = 60×5/18 = 50/3 ≈ 16.67 m/s.

Q037 APTITUDE (TECHNICAL ABILITY)

The
average of the first five odd numbers is

A 5
B 6
C 7
D 9

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
1+3+5+7+9=25.
Average=5.

Q038 APTITUDE (TECHNICAL ABILITY)

12
workers finish a wall in 8 days. How many workers finish it in 6 days?

A 14
B 15
C 16
D 18

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Worker-days
= 12×8=96. Needed = 96/6=16.

Q039 APTITUDE (TECHNICAL ABILITY)

₹ 2,000
at 10% compound interest for 2 years amounts to

A ₹ 2,200
B ₹ 2,400
C ₹ 2,420
D ₹ 2,440

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Amount
= 2000×(1.1)^2 = 2000×1.21 = ₹2,420.

Q040 APTITUDE (TECHNICAL ABILITY)

If
15% of x = 45, then x equals

A 250
B 280
C 300
D 320

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
x
= 45 × 100 / 15 = 300.

Q041 APTITUDE (TECHNICAL ABILITY)

Two cards are drawn from 52 without replacement. Probability both are aces is

A 1/221
B 1/169
C 1/17
D 2/52

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
C(4,2)/C(52,2) = 6/1326 = 1/221.

Q042 APTITUDE (TECHNICAL ABILITY)

[P&C] The number of distinct arrangements of the letters in LEVEL is

A 30
B 60
C 120
D 24

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
LEVEL has 5 letters, L twice and E twice. 5!
/ (2! × 2!) = 30.

Q043 APTITUDE (TECHNICAL ABILITY)

A finishes in 20 days, B in 30 days. Working together they take

A 10 days
B 12 days
C 15 days
D 25 days

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Combined rate = 1/20 + 1/30 = 1/12. Time = 12 days.

Q044 APTITUDE (TECHNICAL ABILITY)

An item marked ₹2,500 is sold after 20% discount. If CP is ₹1,800, profit percent is

A 10%
B 11.11%
C 12%
D 20%

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
SP = 80% of 2500 = 2000. Profit = 200. % = 200/1800 × 100 = 11.11%.

Q045 APTITUDE (TECHNICAL ABILITY)

A and B together take 8 days. A alone takes 12 days. B alone takes

A 18 days
B 20 days
C 24 days
D 30 days

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
B = 1/8 − 1/12 = 1/24. B takes 24 days.

Q046 APTITUDE (TECHNICAL ABILITY)

CP = ₹1,200 and profit = 15%. Selling price is

A ₹1,350
B ₹1,360
C ₹1,380
D ₹1,400

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
SP = 115% of 1200 = ₹1,380.

Q047 APTITUDE (TECHNICAL ABILITY)

A fair coin is tossed three times. Probability of exactly two heads is

A 1/8
B 3/8
C 1/2
D 3/4

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
C(3,2)/8 = 3/8.

Q048 APTITUDE (TECHNICAL ABILITY)

A bag has 3 white and 5 black balls. One ball is drawn. Probability it is black is

A 3/8
B 1/2
C 5/8
D 2/5

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
5 black out of 8 = 5/8.

Q049 APTITUDE (TECHNICAL ABILITY)

How many ways can 5 distinct books be arranged on a shelf if two particular books must stay together?

A 48
B 60
C 120
D 240

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

Tie the two books as one bundle: 4 units can be arranged in 4! ways and the bundle in 2! ways. 24 × 2 = 48.

Q050 APTITUDE (TECHNICAL ABILITY)

From 6 men and 4 women, a committee of 4 with exactly 2 women can be formed in how many ways?

A 60
B 90
C 120
D 180

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

C(4,2) × C(6,2) = 6 × 15 = 90.

Q051 APTITUDE (TECHNICAL ABILITY)

An article is sold at 10% loss for ₹540. The cost price is

A ₹ 580
B ₹ 600
C ₹ 620
D ₹ 650

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

90% of CP = 540, so CP = 540 × 100 / 90 = ₹600.

Q052 APTITUDE (TECHNICAL ABILITY)

Two
coins are tossed. Probability of at least one tail is

A 1/4
B 1/2
C 3/4
D 1

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Outcomes
HH, HT, TH, TT. At least one tail = 3/4.

Q053 APTITUDE (TECHNICAL ABILITY)

12
men finish a job in 10 days. How many men finish the same job in 8 days?

A 14
B 15
C 16
D 18

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

Man-days
= 120. Required men = 120 / 8 = 15.

Q054 APTITUDE (TECHNICAL ABILITY)

A
bag contains 5 green and 3 yellow balls. One ball is drawn at random. Probability it is yellow is

A 3/8
B 5/8
C 3/5
D 1/3

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

3
yellow out of 8 = 3/8.

Q055 APTITUDE (TECHNICAL ABILITY)

How many 3-digit even numbers can be formed from 1, 2, 3, 4, 5 if repetition is not allowed?

A 24
B 30
C 36
D 48

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

The units digit must be 2 or 4. If units=2, tens and hundreds from remaining 4 digits: 4×3=12. Same if units=4. Total 24.

Q056 APTITUDE (TECHNICAL ABILITY)

A chair bought for ₹900 is sold for ₹1,080. The profit percent is

A 15%
B 18%
C 20%
D 25%

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Profit = 180. Percent = 180/900 × 100 = 20%.

Q057 APTITUDE (TECHNICAL ABILITY)

The letters of the word PAPER can be arranged in how many distinct ways?

A 60
B 120
C 24
D 80

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

PAPER has 5 letters and P repeats twice. 5! / 2! = 60.

Q058 APTITUDE (TECHNICAL ABILITY)

A
die is thrown once. What is the probability of a prime number?

A 1/3
B 1/2
C 2/3
D 1/6

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Primes
on a die: 2, 3, 5. Probability = 3/6 = 1/2.

Q059 APTITUDE (TECHNICAL ABILITY)

A
tap fills a tank in 8 hours and a drain empties it in 12 hours. If both are open, the tank fills in

A 16 hours
B 20 hours
C 24 hours
D 48 hours

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Net
work per hour = 1/8 − 1/12 = 1/24. Time = 24 hours.

Q060 APTITUDE (TECHNICAL ABILITY)

In
how many ways can a team of 4 be chosen from 7 players?

A 28
B 30
C 35
D 84

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
C(7,4)
= 35.

Q061 APTITUDE (TECHNICAL ABILITY)

A
fair die is thrown. Probability of a number greater than 4 is

(A) 1/6
(B) 1/3
(C) 1/2
(D) 2/3

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
5
and 6: 2/6 = 1/3.

Q062 APTITUDE (TECHNICAL ABILITY)

A
and B together take 9 days. A alone takes 18 days. B alone takes

(A) 12 days
(B) 15 days
(C) 18 days
(D) 27 days

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
B
= 1/9 − 1/18 = 1/18. B also takes 18 days.

Q063 APTITUDE (TECHNICAL ABILITY)

A
bag bought for ₹480 is sold for ₹552. The profit percent is

(A) 12%
(B) 15%
(C) 18%
(D) 20%

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Profit
= 72. Percent = 72/480 × 100 = 15%.

Q064 APTITUDE (TECHNICAL ABILITY)

How
many 3-digit numbers can be formed from 2, 3, 5, 7, 9 if digits
are not repeated?

A 60
B 120
C 125
D 15

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

P(5,3)
= 5 × 4 × 3 = 60.

Q065 APTITUDE (TECHNICAL ABILITY)

From
8 people, a committee of 3 can be formed in how many ways?

A 24
B 56
C 336
D 512

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

C(8,3)
= 8×7×6 / 6 = 56.

Q066 APTITUDE (TECHNICAL ABILITY)

An
item sold at 20% profit brings ₹720. The cost price is

A ₹ 600
B ₹ 576
C ₹ 640
D ₹ 700

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

120%
of CP = 720, so CP = 720 × 100 / 120 = ₹600.

Q067 APTITUDE (TECHNICAL ABILITY)

A
bag has 5 white and 7 black balls. One ball is drawn. Probability
it is white is

A 5/7
B 1/2
C 5/12
D 7/12

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
5
white out of 12 = 5/12.

Q068 APTITUDE (TECHNICAL ABILITY)

A
finishes a job in 15 days and B in 10 days. Together they take

A 5
days
B 6 days
C 8
days
D 12.5
days

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
Combined
rate = 1/15 + 1/10 = 1/6. Time = 6 days.

Q069 APTITUDE (TECHNICAL ABILITY)

Two
coins are tossed. Probability of two heads is

A 1/4
B 1/2
C 1/3
D 3/4

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
HH
is 1 of 4 outcomes.

Q070 APTITUDE (TECHNICAL ABILITY)

The letters of the word OFFER can be arranged in how many distinct ways?

A 60
B 120
C 24
D 80

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

5 letters with F twice: 5! / 2! = 60.

Q071 APTITUDE (TECHNICAL ABILITY)

A and B together take 15 days. B alone takes 60 days. A alone takes

A 18 days
B 20 days
C 24 days
D 45 days

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

A's rate = 1/15 − 1/60 = 1/20. A takes 20 days.

Q072 APTITUDE (TECHNICAL ABILITY)

The letters of the word ERROR can be arranged in how many distinct ways?

A 20
B 60
C 120
D 24

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

5 letters, R three times. 5! / 3! = 20.

Q073 APTITUDE (TECHNICAL ABILITY)

In
how many ways can 2 students be chosen from 9 students?

A 18
B 36
C 72
D 81

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
C(9,2)=36.

Q074 APTITUDE (TECHNICAL ABILITY)

A
can finish in 24 days and B in 8 days. Together they take

A 6 days
B 8 days
C 10 days
D 12 days

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
1/24+1/8=1/6.
Six days.

Q075 APTITUDE (TECHNICAL ABILITY)

A
card is drawn from 52. Probability it is a queen is

A 1/13
B 1/4
C 1/26
D 4/13

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
4
queens in 52 = 1/13.

Q076 APTITUDE (TECHNICAL ABILITY)

A
number is picked from 1 to 10. Probability it is even is

A 2/5
B 1/2
C 3/5
D 1/5

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
2,4,6,8,10
→ 5/10 = 1/2.

Q077 APTITUDE (TECHNICAL ABILITY)

A
coin is tossed twice. Probability of one head and one tail is

A 1/4
B 1/3
C 1/2
D 3/4

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
HT
and TH: 2/4 = 1/2.

Q078 APTITUDE (TECHNICAL ABILITY)

A
watch sold at 10% profit brings ₹2,200. The cost price is

A ₹ 1,980
B ₹ 2,000
C ₹ 2,100
D ₹ 2,420

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
110%
of CP = 2200, CP = 2000.

Q079 APTITUDE (TECHNICAL ABILITY)

How
many 4-digit numbers can be formed from 1, 2, 3, 4, 5, 6 if digits are not repeated?

A 120
B 240
C 360
D 720

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
P(6,4)=6×5×4×3=360.

Q080 APTITUDE (TECHNICAL ABILITY)

A
lamp bought for ₹750 is sold for ₹900. The profit percent is

A 15%
B 18%
C 20%
D 25%

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Profit
= 150. Percent = 150/750 × 100 = 20%.

Q081 APTTITUDE_TECHNCAL ABILITY

If
a number is divided by 5, remainder = 2. If the same number is divided by 7, remainder = 3. What is the smallest such number?

A 17
B 23
C 38
D 52

CORRECT ANSWER Option (D)

STEP-BY-STEP LOGIC & EXPLANATION:

Number
= 5k + 2. Check divisibility by 7.
k=6
→ 5(6)+2=32 remainder 4.
k=7
→ 37 remainder 2.
k=8
→ 42 remainder 0.
k=9
→ 47 remainder 5.
k=10
→ 52 remainder 3 .
So
number = 52 .

Q082 APTTITUDE_TECHNCAL ABILITY

The
probability of getting a sum of 9 when two dice are thrown is:

A 1/12
B 1/9
C 5/39
D 7/36

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

Combinations:
(3,6), (4,5), (5,4), (6,3).
Total
= 4 outcomes.
Probability
= 4/36 = 1/9 (Correction → 1/9, not 1/12).

Q083 APTTITUDE_TECHNCAL ABILITY

The sum of present ages of A and B is 60. 5 years ago, A was 5 times B. What are their ages?

A. A=55, B=5
B. A=50, B=10
C. A=45, B=15
D. A=40, B=20

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

A+B=60.
(A–5)
= 5(B–5).
Solving
→ A=50, B=10.

Q084 APTTITUDE_TECHNCAL ABILITY

If $x^2 – 5x + 6 = 0$, then find the value of $(1/x_1 + 1/x_2)$.

A. 1/2
B. 5/6
C. 2/3
D. 3/2

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

Roots
= 2, 3.
(1/2
+ 1/3) = (3+2)/6 = 5/6.

Q085 APTTITUDE_TECHNCAL ABILITY

A
can complete a work in 20 days. B is 25% more efficient than A. In how many days will B finish the same work?

A 15
B 16
C 18
D 20

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
A's
1-day work = 1/20.
Efficiency
of B = 125% of A = 5/4 × (1/20) = 1/16.
So
B will finish in 16 days.

Q086 APTTITUDE_TECHNCAL ABILITY

A
shopkeeper allows 20% discount on the marked price and still makes 25% profit. If the marked price is ₹1000, what is the cost price?

A ₹ 600
B ₹ 625
C ₹ 650
D ₹ 700

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
SP
= 1000 – 20% of 1000 = 800.
Profit
= 25%.
CP
= 800 / 1.25 = 640 (correction, not 600).

Q087 APTTITUDE_TECHNCAL ABILITY

The ratio of investments of A and B is 4:5. A gets ₹400 less than B from the profit. Find the total profit.

A ₹ 3600
B ₹ 4000
C ₹ 4200
D ₹ 4800

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

Difference = 1 part.
1 part = 400.
Total = 9 parts = 9×400=3600.

Q088 APTTITUDE_TECHNCAL ABILITY

A sum of ₹10,000 is invested at 20% per annum compounded annually. What will be the compound interest after 2 years?

A ₹ 4000
B ₹ 4200
C ₹ 4400
D ₹ 4800

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

CI = P[(1 + r/100)^t – 1] = 10000[(1.2)^2 – 1] = 10000(1.44 – 1) = 4400.
(Answer : ₹4400).

Q089 APTTITUDE_TECHNCAL ABILITY

The
average of 5 consecutive odd numbers is 35. What is the largest of these numbers?

A 37
B 39
C 41
D 43

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

Middle
term = 35 (average of consecutive odd nos).
Series
= 31, 33, 35, 37, 39 → largest = 39.

Q090 APTTITUDE_TECHNCAL ABILITY

Two
pipes can fill a tank in 15 min and 20 min respectively. If both pipes are opened together, how long will it take to fill the tank?

A 7 min
B 8 min
C 8 min 34 sec
D 9 min

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

1/15
+ 1/20 = (4+3)/60 = 7/60.
Time
= 60/7 = 8.57 min ≈ 8 min 34 sec.